

NUMERICAL EXAMPLES

1 ✓ **EXAMPLE 5** Two straight and narrow parallel slits 1 mm apart are illuminated by monochromatic light. Fringes formed on the screen held at a distance of 100 cm from the slits are 0.50 mm apart. What is the wavelength of light?

Solution : According to the problem $\beta = 0.50$ mm. $= 0.05$ cm, $2d = 1$ mm $= 0.1$ cm, and $D = 100$ cm.

The fringe width is given by

$$\beta = \frac{\lambda D}{2d} \text{ or } \lambda = \frac{\beta \times 2d}{D}$$

Substituting the given values, we have

$$\begin{aligned} \lambda &= \frac{0.005 \times 0.1}{100} = 5 \times 10^{-5} \text{ cm} \\ &= 5000 \text{ \AA}. \end{aligned}$$

2 ✓ **EXAMPLE 6** Two coherent sources are placed 0.9 mm apart and the fringes are observed one metre away. Calculate the wavelength of the monochromatic light used if it produces the second dark fringe at a distance of 10 mm from the central fringe.

Solution : Here $2d = 0.9$ mm $= 0.09$ cm, $D = 1$ metre $= 100$ cm, $n = 2$, $x_2 = 10$ mm $= 1$ cm.

For a dark fringe.

$$x_n = \frac{D}{2d} (2n + 1) \frac{\lambda}{2} \text{ or } \lambda = \frac{4d}{D} \frac{x_n}{(2n + 1)}$$

Substituting the given values, we have

$$\lambda = \frac{0.09 \times 2 \times 1}{100 \times 5} = 3.6 \times 10^{-5} \text{ cm}.$$

3 ✓ **EXAMPLE 7** A biprism is placed 5 cm from a slit illuminated by sodium light ($\lambda = 5890 \text{ \AA}$). The width of the fringes obtained on a screen 75 cm from the biprism is 9.424×10^{-2} cm. What is the distance between the two coherent sources?

Solution : Here $\lambda = 5890 \text{ \AA} = 5890 \times 10^{-8}$ cm,

$\beta = 9.424 \times 10^{-2}$ cm, $D = 5 + 75 = 80$ cm, $2d = ?$

We know that $\beta = \frac{\lambda D}{2d}$

26.9

$$\therefore 2d = \frac{\lambda D}{\beta} = \frac{5890 \times 10^{-8} \times 80}{9.424 \times 10^{-2}} = 0.05 \text{ cm}$$

4
EXAMPLE 16 A thin sheet of glass ($\mu = 1.5$) of 6 microns thickness introduced in the path of interfering beams in a biprism arrangement shifts the central fringe to a position normally occupied by the fifth. Find the wavelength of light used. (1 micron = 10^{-4} cm).

Solution : Here $\mu = 1.5$, $t = 6$ microns = 6×10^{-4} cm, $S = 5\beta$.

The thickness t of the glass plate is given by

$$t = \frac{S\lambda}{(\mu - 1)\beta} \text{ or } \lambda = \frac{t(\mu - 1)\beta}{S}$$

$$\begin{aligned} \therefore \lambda &= \frac{6 \times 10^{-4} \times (1.5 - 1)\beta}{5\beta} \\ &= 6 \times 10^{-5} = 6000 \times 10^{-8} \text{ cm} \\ &= 6000 \text{ \AA} \end{aligned}$$

5
EXAMPLE 17 The central fringe of the interference produced by light of wavelength $6000 \text{ \AA}$ is shifted to the position of 5th bright fringe by introducing a thin glass plate of refractive index 1.5. Calculate the thickness of the plate.

Solution : We know that the shift S of the central or of any other fringe is given by

$$S = \frac{D}{2d} (\mu - 1)t = \frac{\beta}{\lambda} (\mu - 1)t$$

where β is the fringe width.

According to the given problem.

$$S = 5\beta, \lambda = 6000 \text{ \AA} = 6 \times 10^{-5} \text{ cm and } \mu = 1.5.$$

$$\therefore 5\beta = \frac{\beta(1.5 - 1)t}{6 \times 10^{-5}}$$

6 or $t = \frac{5 \times 6 \times 10^{-5}}{0.5} = 6 \times 10^{-4} \text{ cm.} =$

EXAMPLE 18 In a double slit arrangement, fringes are produced using light of wavelength $4800 \text{ \AA}$. One slit is covered by a thin plate of glass of refractive index 1.4

and the other slit by another plate of glass of the thickness but of refractive index 1.7. On doing so the central bright fringe shifts to the position originally occupied by the fifth bright fringe from the centre. Find the thickness of the glass plate.

Solution : We know that

$$S = \frac{\beta}{\lambda} (\mu - 1) t$$

If another plate of the same thickness t but refractive index μ' is introduced in the other beam, then shift becomes

$$S = \frac{\beta}{\lambda} (\mu - \mu') t$$

Here $S = 5\beta$, $\lambda = 4.8 \times 10^{-5}$ cm

and $\mu - \mu' = 0.3$

$$\therefore 5\beta = \frac{\beta (0.3)}{4.8 \times 10^{-5}} t$$

$$\text{or } t = \frac{5 \times 4.8 \times 10^{-5}}{0.3} = 8.0 \times 10^{-4} \text{ cm.}$$

EXAMPLE 28 In a Newton's ring experiment, the diameter of the 5th ring was 0.3 cm and the diameter of 25th ring was 0.8 cm. If the radius of curvature of the plano convex lens is 100 cm, find the wavelength of light used.

Solution : We know that $\lambda = \frac{D_{n+p}^2 - D_n^2}{4 p R}$

Here $D_{25} = 0.8$ cm, $D_5 = 0.3$ cm.

$$p = 25 - 5 = 20, \text{ and } R = 100 \text{ cm}$$

Substituting these values, we have

$$\lambda = \frac{(0.8)^2 - (0.3)^2}{4 \times 20 \times 100}$$

$$= 4.87 \times 10^{-5} \text{ cm. } 6.875 \times 10^{-5}$$

$$= 6875 \text{ \AA}$$

EXAMPLE 29 A Newton's ring arrangement is used with a source emitting two wavelengths $\lambda_1 = 6 \times 10^{-5}$ cm and 4.5×10^{-5} cm and it is found that n th dark ring due to λ_1 coincides with $(n+1)$ th dark ring for λ_2 . If radius of curvature of the curved surface is 90 cm, find the diameter of n th dark ring for λ_1 .

Solution : The diameter of n th dark ring due to λ_1 is given by

$$[D_n^2]_{\lambda_1} = 4nR\lambda_1$$

The diameter of $(n+1)$ th dark ring due to λ_2 is given by

$$[D_{n+1}^2]_{\lambda_2} = 4(n+1)R\lambda_2$$

According to the problem

$$[D_n^2]_{\lambda_1} = [D_{n+1}^2]_{\lambda_2}$$

$$\therefore 4nR\lambda_1 = 4(n+1)R\lambda_2$$

$$n\lambda_1 = (n+1)\lambda_2$$

$$\text{or } n = \frac{\lambda_2}{\lambda_1 - \lambda_2} = \frac{4.5 \times 10^{-5}}{(6.0 - 4.5) \times 10^{-5}} = 3$$

$$\begin{aligned} \text{Now } [D_n^2]_{\lambda_1} &= 4 \times 3R\lambda_1 \\ &= 4 \times 3 \times 90 \times 6 \times 10^{-5} \\ [D_n]_{\lambda_1} &= \sqrt{4 \times 3 \times 90 \times 6 \times 10^{-5}} \\ &= 0.254 \text{ cm.} \end{aligned}$$

EXAMPLE 30 In Newton's ring experiment, the diameters of the 4th and 12th dark rings are 0.400 cm and 0.700 cm respectively. Find the diameter of the 20th dark ring.

Solution : We know that

$$D_{n+p}^2 - D_n^2 = 4p\lambda R$$

Here $(n+p) = 12, n = 4$, i.e., $p = 12 - 4 = 8$,

$D_{n+p} = 0.700$ cm, and $D_n = 0.400$ cm.

$$\therefore D_{12}^2 - D_4^2 = 4 \times 8 \times \lambda R$$

For 20th and 4th rings

$$D_{20}^2 - D_4^2 = 4 \times 16 \times \lambda R$$

$$[\because p = 20 - 4 = 16] \quad \dots(2)$$

Dividing (2) by (1)

$$\frac{D_{20}^2 - D_4^2}{D_{12}^2 - D_4^2} = \frac{4 \times 16 \times \lambda R}{4 \times 8 \times \lambda R} = 2$$

$$\text{or } \frac{D_{20}^2 - (0.4)^2}{(0.7)^2 - (0.4)^2} = 2 \quad \text{or } \frac{D_{20}^2 - 0.16}{1.1 \times 0.3} = 2$$

$$\text{or } D_{20}^2 - 0.16 = 1.1 \times 0.3 \times 2 = 0.66$$

$$D_{20}^2 = 0.66 + 0.16 = 0.82 \quad \text{or } D_{20} = 0.906 \text{ cm.}$$

EXAMPLE 31 Newton's rings formed by sodium light between a flat glass plate and a convex lens are viewed normally. What will be the order of the dark ring which will have double the diameter of that of 40th dark ring.

Solution : We know that $D_n^2 = 4n\lambda R$

According to the given problem,

$$D_{40}^2 = 4 \times 40 \times \lambda R \quad \dots(1)$$

Let n be the order of dark ring whose diameter is twice that of the diameter of 40th dark ring, i.e., for this dark ring diameter is $(2D_{40})$. Hence

$$(2D_{40})^2 = 4n\lambda R \quad \dots(2)$$

From equations (1) and (2)

$$\frac{(2D_{40})^2}{(D_{40})^2} = \frac{n}{40} \quad \text{or } n = 40 \times 4 = 160$$

EXAMPLE 32 In a Newton's rings experiment the diameter of the 10th ring changes from 1.40 to 1.27 cm when a liquid is introduced between the lens and the plate. Calculate the refractive index of the liquid.

Solution : When the liquid is used, the diameter of the 10th ring is given by

$$(D_{10})^2 = \frac{4 \times 10 \lambda R}{\mu}$$

For air medium $(D_{10})^2 = 4 \times 10 \lambda R$

$$\therefore \mu = \frac{D_{10}^2}{(D'_{10})^2} = \frac{(1.40)^2}{(1.27)^2} = 1.215$$

EXAMPLE 33 Light containing two wavelengths λ_1 and λ_2 falls normally on a planoconvex lens of radius of curvature R resting on a glass plate. If the n th